IoT-Based Predictive Maintenance for Electric Motors

Agyei Festus Kusi
Opoku Pearl Adjei
Amoah Bismark Kyei

Supervised By: Dr. Vincent M. Nofong

University of Mines and Technology
Faculty of Computing and Mathematical Science
Computer Science and Engineering Department

March ,2024

Presentation Outline

1. Introduction
2. Statement of Problem
3. Objectives
4. Tools and Facilities to be Used
5. Methods to be Used
6. Scope of Work
7. Expected Outcomes
8. Conclusion
9. References

Introduction

- Industries that rely on electric motors often experience costly unplanned failures due to inefficient maintenance strategies(Lee et al., 2019).
- **Reactive maintenance** leads to unexpected breakdowns, while **Preventive maintenance** wastes resources by following fixed schedules(Kusiak, 2017).

Statement of Problem

- Existing monitoring systems provide real-time data but fail to predict faults, leading to ineffective maintenance decisions.
- The absence of **intelligent diagnostics** makes it difficult to classify and address motor failures before they occur.
- Hence the need for a **predictive maintenance system** that integrates IoT, machine learning, and intelligent diagnostics to enhance motor **reliability** and **efficiency**.

Objectives:

- Develop an IoT-based system for real-time monitoring of electric motors using smart sensors.
- Implement machine learning models to classify motor faults and predict failures.
- Develop an intelligent chatbot to assist maintenance teams with automated diagnostics and troubleshooting.
- Create an automated reporting system to provide maintenance insights and performance analytics.

Hardware Tools:

- Sensors(Temperature, Vibration,Current,Voltage)
- Microcontroller (Arduino Uno)
- GSM Module

Software Tools:

- ThingSpeak
- Node-RED.
- Python (TensorFlow, Scikit-learn, NumPy)
- Power BI / Grafana
- GPT-based chatbot

Methods to be Used

- Extensive review of relevant literature and analysis
- Collaborative consultations with supervisor and faculty

Scope of Work

This research focuses on IoT-based predictive maintenance for electric motors, integrating real-time monitoring, fault classification, and energy consumption analysis. It aims to reduce unplanned downtime, optimize power usage, and enhance fault diagnostics through machine learning and intelligent automation, ensuring improved reliability and efficiency in industrial applications.

Expected Outcomes

- A fully functional IoT-based predictive maintenance system for electric motors.
- Improved energy efficiency, scalability, and reliability.
- Improved fault detection and classification accuracy using machine learning.

Conclusion

This research develops an **IoT-based predictive maintenance system for electric motors**, **integrating real-time monitoring**, **fault classification**, and **energy optimization**. Using machine learning and intelligent diagnostics, it aims to prevent failures, lower costs, and enhance efficiency, providing a scalable, cost-effective alternative to traditional maintenance methods.

References

- Kusiak, A (2017),"Smart manufacturing must embrace big data", *Nature*, Vol.544, No.7648, pp.23–25.
- Lee, J., Kao, H. and Yang, S (2019),"Service innovation and smart analytics for Industry 4.0 and big data environment", *Procedia CIRP*, Vol.16, pp.3–8.

Thank You!!!